GenAI / LLMs

with Java 21

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GenAI

Generative artificial intelligence (**generative AI**, **GenAI**,[1] or **GAI**) is artificial intelligence capable of generating text, images, videos, or other data using generative models,[2] often in response to prompts.[3][4] Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.[5][6]

Improvements in transformer-based deep neural networks, particularly large language models (LLMs), enabled an AI boom of generative AI systems in the early 2020s. These include chatbots such as ChatGPT, Copilot, Gemini and LLaMA, text-to-image artificial intelligence image generation systems such as Stable Diffusion, Midjourney and DALL-E, and text-to-video AI generators such as Sora.[7][8][9][10] Companies such as OpenAI, Anthropic, Microsoft, Google, and Baidu as well as numerous smaller firms have developed generative AI models.[3][11][12]

https://en.wikipedia.org/wiki/Generative_artificial_intelligence

Transformer

A **transformer** is a deep learning architecture developed by Google and based on the multi-head attention mechanism, proposed in a 2017 paper "Attention Is All You Need".[1] Text is converted to numerical representations called tokens, and each token is converted into a vector via looking up from a word embedding table.[1] At each layer, each token is then contextualized within the scope of the context window with other (unmasked) tokens via a parallel multi-head attention mechanism allowing the signal for key tokens to be amplified and less important tokens to be diminished. The transformer paper, published in 2017, is based on the softmax-based attention mechanism proposed by Bahdanau *et. al.* in 2014 for machine translation,[2][3] and the Fast Weight Controller, similar to a transformer, proposed in 1992.[4][5][6]

https://en.wikipedia.org/wiki/Transformer_(deep_learning_architecture)

Token

- AWS Bedrock: Use 6 characters per token as an approximation for the number of tokens.
- https://platform.openai.com/tokenizer

RAG

RAG

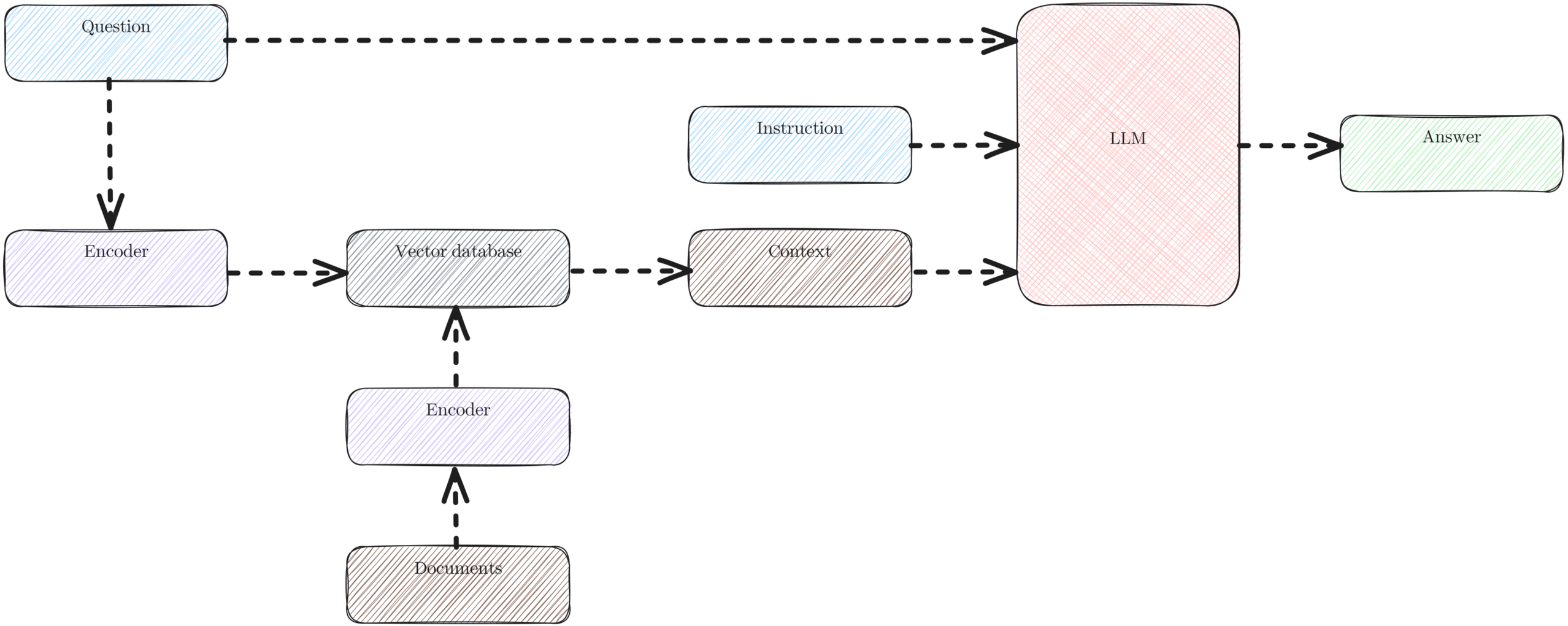
Retrieval-augmented generation [edit]

Retrieval-Augmented Generation (RAG) is a two-phase process involving document retrieval and answer formulation by a Large Language Model (LLM). The initial phase utilizes dense embeddings to retrieve documents. This retrieval can be based on a variety of database formats depending on the use case, such as a vector database, summary index, tree index, or keyword table index.[51]

In response to a query, a document retriever selects the most relevant documents. This relevance is typically determined by first encoding both the query and the documents into vectors, then identifying documents whose vectors are closest in Euclidean distance to the query vector. Following document retrieval, the LLM generates an output that incorporates information from both the query and the retrieved documents.[52] This method is particularly beneficial for handling proprietary or dynamic information that was not included in the initial training or fine-tuning phases of the model. RAG is also notable for its use of "few-shot" learning, where the model uses a small number of examples, often automatically retrieved from a database, to inform its outputs.

https://en.wikipedia.org/wiki/Prompt_engineering

RAG



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Vector Database

Vector Database

A **vector database**, **vector store** or **vector search engine** is a database that can store vectors (fixed-length lists of numbers) along with other data items. Vector databases typically implement one or more Approximate Nearest Neighbor (ANN) algorithms,[1][2] so that one can search the database with a query vector to retrieve the closest matching database records.

Vectors are mathematical representations of data in a high-dimensional space. In this space, each dimension corresponds to a feature of the data, with the number of dimensions ranging from a few hundred to tens of thousands, depending on the complexity of the data being represented. A vector's position in this space represents its characteristics. Words, phrases, or entire documents, as well as images, audio, and other types of data, can all be vectorized.[3]

These feature vectors may be computed from the raw data using machine learning methods such as feature extraction algorithms, word embeddings[4] or deep learning networks. The goal is that semantically similar data items receive feature vectors close to each other.

Vector databases can be used for similarity search, multi-modal search, recommendations engines, large language models (LLMs), etc.[5]

https://en.wikipedia.org/wiki/Vector_database

Vector Database (contd.)

Vector databases are also often used to implement Retrieval-Augmented Generation (RAG), a method to improve domain-specific responses of large language models. The retrieval component of a RAG can be any search system, but is most often implemented as a vector database. Text documents describing the domain of interest are collected, and for each document or document section, a feature vector (known as an "embedding") is computed, typically using a deep learning network, and stored in a vector database. Given a user prompt, the feature vector of the prompt is computed, and the database is queried to retrieve the most relevant documents. These are then automatically added into the context window of the large language model, and the large language model proceeds to create a response to the prompt given this context.[6]

https://en.wikipedia.org/wiki/Prompt_engineering

LLMs

- ChatGPT
- Claude
- Llama
- Mistral

LangChain4j

- similar to multiple EntityManagers behind a boundary
- llm abstraction similar to JPA
- fully integrated with MicroProfile / Quarkus
- works with GraalVM

LangChain4j

- prompts
- chains
- agents and tools
- memory
- llm
- indexes
- DocumentSplitter, VectorDB, DocumentLoader